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Original article

Associations Between Labor Control Through Digital Platforms and Workers' Mental Wellbeing: A Survey of Location-based Platform Workers in Taiwan

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ABSTRACT

Background: Platform work offers flexibility and autonomy to workers, but there are concerns about the impacts of excessive labor control exercised by digital platforms using algorithmic management. This study assessed the level of labor control exerted by digital platforms and explored its associations with psychosocial work conditions and workers' mental well-being.

Methods: A total of 487 location-based platform workers in Taiwan, encompassing ride-hailing, delivery, and house chore services, participated in this questionnaire survey. A 5-item scale was created to assess platform-mediated labor control. Psychosocial work conditions, including demands, job control, work rewards, and mental well-being, including burnout and self-rated mental health status, were assessed using validated instruments. Cronbach's α and principal component analysis were employed to assess the validity of the labor control scale. Logistic regression analysis and path analysis were conducted to investigate the relationship between labor control and workers' mental health.

Results: Exploratory factor analyses confirmed the structural validity and the internal consistency of the scale. Logistic regression models revealed that higher levels of labor control were associated with an increased risk of burnout and poorer mental health. Path analyses further indicated that higher labor control through digital platforms raised mental health risks by increasing job demands.

Conclusion: Algorithmic management is associated with increased stress among platform workers as they try to meet the platform's performance metrics.

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1. Introduction

Innovation in information and communication technology (ICT) has triggered a swift expansion of work conducted through online platforms across the globe. In several countries, platform work is viewed favorably as a tool to stimulate economic growth and enhance employment opportunities. It addresses labor shortages for businesses while providing flexibility for both businesses and workers. However, apprehensions persist regarding the precarious nature of platform jobs, which often involve remuneration based on piece rates with low pay. These jobs are characterized by poor job quality, and limited career prospects, and are generally excluded from the coverage of labor and social protection systems [1]. A substantial body of literature indicates that the advent of platform work introduces several challenges to conventional employment practices, work quality, and the safety and well-being

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of workers. Moreover, it has the potential to worsen existing social inequalities in both work and health conditions [2–5].

Digital labor platforms can be broadly categorized into two major types. The first type is "location-based" platforms, which mediate work online but allocate tasks to workers based on a specific geographical area. These platforms typically involve local and service-oriented tasks, such as driving, house cleaning, or running errands within a designated location. The other is "web-based" platforms, where businesses and clients can outsource work tasks to a crowd of workers dispersed across different locations [6]. In both types of platforms, algorithmic techniques play a central role in the functioning of platform work, and the extent of workers' autonomy in making workplace decisions is considered a key factor affecting their satisfaction and well-being in platform work [7].

One of the prominent work hazards at the organizational level is the level of labor control exerted by digital labor platforms [8]. Platforms utilize sophisticated algorithmic techniques to allocate and control work assignments, set prices, enforce or penalize certain practices, and monitor and reward workers' performance in a real-time manner [9]. However, the extent to which platforms exert their labor control power and its impact on workers' mental health has been rarely examined. With the use of surveillance tools and algorithmic management techniques, platforms not only function as employers in practice but also wield significant control to the extent that workers' autonomy and meaningfulness of their work are undermined. Some studies have delved into how digital platform exerted labor control. For instance, Griesback and colleagues documented how Canadian food delivery workers' job autonomy over working time and activities was restricted by the platforms [10]. In another study, Wu et al examined the influence of labor control among Uber drivers in China. They identified three mechanisms through which Uber controlled drivers' working behaviors: an incentive pay system, an evaluation system, and an arbitrary system to determine work arrangements [11]. Furthermore, there exists inherent information asymmetry and power imbalance between the platforms and the workers. As indicated by previous studies, workers subjected to constant monitoring and evaluation by digital algorithms may experience reduced job autonomy and an increased risk of psychological distress [12,13]. However, to the best of our knowledge, only a limited number of studies have quantified the extent of labor control and examined its influences on work characteristics and workers' mental health outcomes.

The existing literature also highlights numerous psychosocial work hazards that are specific to platform work. These hazards include the blurring of work-life boundaries, the uncertainty surrounding employment and income, intense market competitions, alienating work tasks, social isolation, limited career prospects, cyberbullying, and techno-stress induced by heavy reliance on ICT at work. Issues such as extra time spent searching on the internet for tasks, taking unpaid qualification tests, researching clients, and responding to online comments are also mentioned in some studies as sources of psychological stress [6,14,15]. As performance-based and piece-rated pay systems have been identified to increase risks of overwork and stress-related health problems in previous studies [5,16], it is reasonable to anticipate that platform workers performing piece-rated work tasks may encounter similar issues.

In Taiwan, the platform economy is flourishing, with ride-hailing drivers and delivery motorbike riders being the most visible among various types of platform work. However, other forms of platform work are also experiencing growth. Unfortunately, official statistics on the scale and workforce size of the platform economy are currently unavailable. This lack of data underscores the importance of conducting further research and data collection to gain a comprehensive understanding of the extent and implications of platform work in the country. In October 2019, there were two unfortunate incidents where motorbike riders died while delivering food. These incidents garnered significant public attention, and the platforms' response to these casualties further intensified public outrage. The platforms claimed that these two riders were considered contractors rather than employees, thereby evading responsibilities for providing workers' compensation benefits. This stance taken by the platforms fueled public anger and raised concerns about the treatment and protection of platform workers in such situations. Due to amounting criticisms from labor rights groups, the Ministry of Labor (MOL) launched an investigation into the practices of all food delivery platform companies. In response to the controversies surrounding the uncertain employment status of food delivery workers, the MOL issued guidance for recognizing employment relationships in November 2019 [17]. Additionally, a checklist for assessing the level of labor subordination (details explained in Section 2.3) was provided, comprising 25 items for employers, workers, and labor inspection agencies to consider in resolving similar disputes. Subsequently, additional regulations and guidelines concerning the occupational safety and health risks of platform work were developed. However, it is noteworthy that these regulations have predominantly centered around food delivery platform work. As far as we are aware, there have been few studies that have explored and compared workers' experiences while engaging in different types of platform work.

In this study, we assessed the level of labor control exerted by the digital platform and investigated its associations with psychosocial work characteristics and workers' mental well-being among location-based platform workers in Taiwan. In this study, mental well-being was assessed using the Brief Symptom Rating Scale (BSRS-5) and burnout was measured using the Copenhagen Burnout Inventory (CBI). The BSRS-5 evaluates mood symptoms and identifies minor mental disorder with good sensitivity and specificity [18],while the CBI focused on fatigue and exhaustion experiences related to work conditions [19]. The hypothesis is that the interplay of varying levels of job demand, job control, work rewards, and platform work might be linked to workers' psychological distress. The study framework is depicted in Fig. 1.

2. Materials and methods

2.1. Study design

An online survey was conducted to understand the socioeconomic profiles, work motives, and work and health conditions of platform workers from the three most visible types of location-based platform work (i.e., ride-hailing drivers, delivery workers, and chore service workers). This study was a part of a research project designed to examine the influence of the platform economy on psychosocial work quality and occupational safety and health risks in Taiwan. The questionnaire was designed based on qualitative information obtained in the initial stage of this project. This study was reviewed and approved by the Research Ethics Committee of the National Taiwan University (No: 202006HS009).

2.2. Study participants and data collection procedure

Eligible participants for the survey were workers aged between 20 and 65 years old who had provided services through digital labor platforms (including online platforms, Facebook communities, and social media associations) within the three months preceding the survey.

This study employed convenience sampling. To recruit participants, a survey link was disseminated through various channels, including bulletin boards, Facebook communities, Line groups of Uber

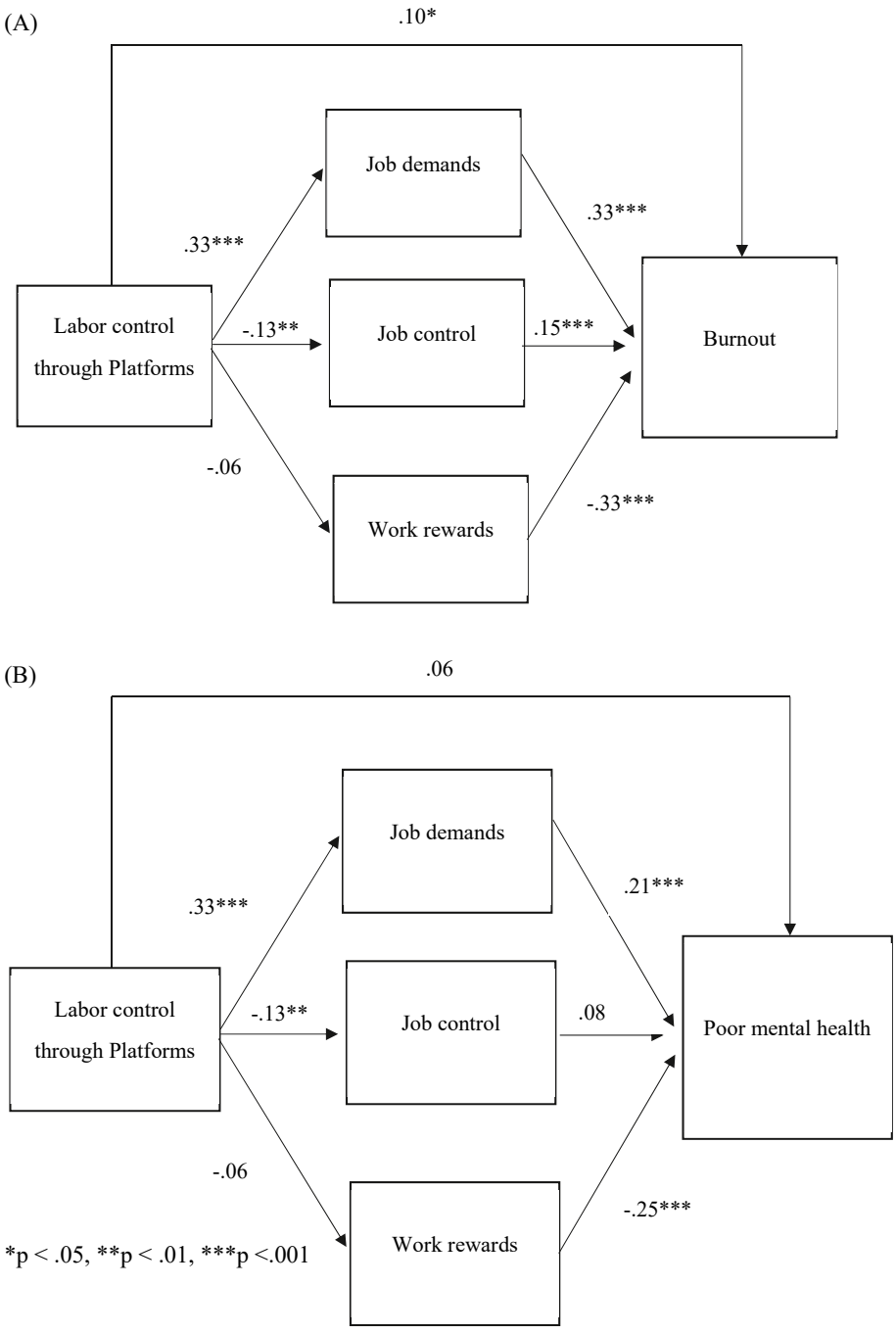


Fig. 1. Path analysis results for (A) burnout (B) poor mental health ($n = 487$).

drivers, food and parcel delivery workers, and cleaners, as well as through personal networks. Additionally, an advertisement was posted on the bulletin board of Taiwan's largest chore-work platform (https://www.chickpt.com.tw/). Participants were informed of the purpose and details of this study before answering the questionnaire. The primary aim of this study was to assess the reliability and validity of a 5-item scale measuring the level of labor control experienced by workers through digital platforms. Due to limited data available on the extent of labor control among platform workers and the overall size of this workforce in Taiwan, we referred to a previous psychometric study and estimated a sample size of 400 participants [20].

Data were collected anonymously through the SurveyCake platform. Information on participants' sociodemographic characteristics, types of platform work, work motives, working conditions, self-rated health, and psychological distress was gathered. Participants were asked to provide their email addresses to receive a digital coupon as a token of appreciation for their participation. A total of 493 workers participated and completed the questionnaire from 12/16/2021 to 01/13/2022. However, five cases had missing values for sex or platform type and were therefore excluded from the analysis, resulting in 488 valid observations.

Table 1
Sociodemographic characteristics, work conditions, and mental health of location-based platform workers

	All types (n = 487)		Ride-hailing drivers (n = 58, 11.91%)		Delivery workers (n = 293, 60.16%)		Chore service workers (n = 136, 27.93%)		p
	n/mean	(%)/SD	n/mean	(%)/SD	n/mean	(%)/SD	n/mean	(%)/SD	
Gender									***
Male	276	(56.67)	54	(93.10)	179	(61.09)	43	(31.62)	
Female	211	(43.33)	4	(6.90)	114	(39.91)	93	(68.38)	
Age (year)	35.39	9.23	39.71	9.13	35.66	8.14	32.96	10.67	***
Labor control	59.41	21.51	58.85	23.53	63.57	19.82	50.71	21.62	***
Low	138	(28.34)	16	(27.59)	58	(19.80)	64	(47.06)	
Median	245	(50.31)	31	(53.45)	155	(52.90)	59	(43.38)	
High	104	(21.36)	11	(18.97)	80	(27.30)	13	(9.56)	
Job demands	64.82	17.62	61.26	20.38	69.03	15.86	57.25	17.20	***
Low	134	(27.52)	20	(34.48)	50	(17.06)	64	(47.06)	
Median	209	(42.92)	22	(37.93)	136	(46.42)	51	(37.50)	
High	144	(29.57)	16	(27.59)	107	(36.52)	21	(15.44)	
Job control	46.15	22.58	45.02	22.48	41.83	22.58	55.92	19.54	***
Low	107	(21.97)	12	(20.69)	86	(29.35)	9	(6.62)	
Median	224	(46.00)	30	(51.72)	127	(43.34)	67	(49.26)	
High	156	(32.03)	16	(27.59)	80	(27.30)	60	(44.12)	
Work rewards	38.80	22.18	40.52	21.71	33.51	22.43	49.45	17.57	***
Low	154	(31.62)	16	(27.59)	123	(41.98)	15	(11.03)	
Median	153	(31.42)	21	(36.21)	89	(30.38)	43	(31.62)	
High	180	(36.96)	21	(36.21)	81	(27.65)	78	(57.35)	
Burnout									**
<50	284	(58.32)	37	(63.79)	151	(51.54)	96	(70.59)	
50–74	156	(32.03)	16	(27.59)	106	(36.18)	34	(25.00)	
≥75	47	(9.65)	5	(8.62)	36	(12.29)	6	(4.41)	
Mental health score (BSRS-5)†									*
<6	266	(54.62)	32	(55.17)	144	(49.15)	90	(66.18)	
6–9	113	(23.20)	17	(29.31)	75	(25.60)	21	(15.44)	
10–14	77	(15.81)	5	(8.62)	54	(18.43)	18	(13.24)	
≥15	31	(6.37)	4	(6.90)	20	(6.83)	7	(5.15)	

*$p < 0.05$; **$p < 0.01$; ***$p < 0.001$ in chi-square tests or ANOVA tests.
†BSRS, Brief symptom rating scale.

2.3. The scale of labor control

We developed a scale for labor control based on the "Guidelines for Identification of Labor Subordination" and its attachment, the "Checklist for Subordination of Labor Contract," which was promulgate d by the Ministry of Labor, Taiwan in 2019 [21]. In the Checklist, personal subordination was defined as the state where a worker must obey the management and face punishment for any wrongdoings. The Checklist contained several items for assessment to measure the level of subordination. The higher the number of checked items, the higher the level of subordination. The following five items were selected from the Checklist to assess the level of labor control among platform workers: (1) at work, my working time is subject to the command or supervision of the platform, (2) at work, the way I work is subject to the command or supervision of the platform, (3) at work, the place where my work is performed is under the command or supervision of the platform, (4) I must abide by the work rules set by the platform; otherwise, I will be punished by the platform, and (5) at work, my work performance is monitored or evaluated by the platform. All five items were rated on a 4-point Likert scale, ranging from 1 (strongly disagree) to 4 (strongly agree). The summary score was standardized, with a higher value indicating a stronger level of labor control. Subsequently, the scores were categorized into tertiles, creating low, medium, and high groups based on the level of labor control.

2.4. Measurement of work characteristics: job demands, job control, and work rewards

The Chinese version of the Job Content Questionnaire (C-JCQ), based on Karasek's Demand-Control model, was utilized to assess psychosocial job demands and control [22]. Five-items were selected from the original scale of job demands and job control to better capture the nature of platform work. Items related to insufficient manpower, as well as two reverse-coded items ("my workloads are not excessive" and "I have enough time to get the job done") were omitted from the original job demands scale. As a result, the scale of job demands comprised five items, namely fast work, hard work, physically demanding work, work requiring concentration, and hectic work. Meanwhile, the scale of job control consisted of 5 items, including learning new things, creative work, work involving multiple tasks, opinions influential, and developing one's abilities. For the sake of brevity, four items were removed from the original scale that was repetitive or less relevant to platform work: "repetitive tasks," "requiring skills", "don't get to decide on how I do my work" (reverse-coded), and "allowed to make my own decision." It has been demonstrated that the measurement of job control and demands with a subset of items remains valid [23].

Additionally, we incorporated the work rewards component of the Siegrist Effort-Reward Imbalance model and constructed a scale of work rewards with four items [24]. The four items in the scale were job security, career prospects, help and support, and appropriate income. Participants rated all items on a 4-point Likert scale, ranging from 1 (strongly disagree) to 4 (strongly agree). For each scale, a summarized score was standardized and subsequently divided into tertiles, creating low, medium, and high groups based on the level of work rewards.

Participants reported various work status characteristics, including self-reported work tenure, income from platform work, work motives, and self-perceived employment status. Additionally, other work characteristics were also assessed, such as working hours, respect from others, job insecurity as well as experiences of work-related injuries and illnesses over the past year.

To maximize the utilization of all data, the mean imputation method was employed to fill in missing value for work-related scales. For each scale, the missing value was imputed with the

Table 2
Internal consistency of scales of work characteristics

Scale	Item	Cronbach's α after deleting the item	Cronbach's α
Labor control through platforms ($n = 471$)*	1. At work, my working time is subject to the command or supervision of the platform.	0.85	0.86
	2. At work, the way I work is subject to the command or supervision of the platform.	0.81	
	3. At work, the place where my work is performed is under the command or supervision of the platform.	0.82	
	4. I must abide by the work rules set by the platform, otherwise, I will be punished by the platform.	0.87	
	5. At work, my work performance is monitored or evaluated by the platform.	0.84	
Job demands ($n = 485$)*	1. My work requires me to work fast.	0.83	0.80
	2. My work requires me to work hard.	0.71	
	3. My work requires a lot of physical effort.	0.74	
	4. My work requires me to concentrate on the job for a long time.	0.78	
	5. I am very hectic at work.	0.75	
Job control ($n = 478$)*	1. My work requires me to learn new things.	0.84	0.86
	2. My work requires me to be creative	0.82	
	3. I can do a variety of tasks on my job.	0.83	
	4. On things that happen at work, my opinions are influential.	0.82	
	5. My work provides room to develop abilities.	0.82	
Work rewards ($n = 480$)*	1. My job is secure.	0.75	0.83
	2. My prospects for career development and promotions are good.	0.76	
	3. When encountering difficulties at work, I receive appropriate help and support	0.79	
	4. My salary or work income are appropriate.	0.84	

*Due to missing data, number of observations for each scale is not consistent.

average response of the other items within the same scale. However, if the number of items with missing values in a scale equaled or exceeded half of the total items in that scale, the entire observation was treated as missing for that particular scale. At this stage, one participant with missing values on the job control scale was removed from the dataset, resulting 487 participants for data analyses.

2.5. Measurements of mental well-being

Two indicators to measure mental well-being were used: psychological symptoms assessed by the BSRS-5 and burnout measured by the Copenhagen Burnout Inventory. The BSRS-5 was captured psychological symptoms experienced over the past week [18,25]. The five items included trouble sleeping, feeling tense or keyed up, feeling easily annoyed or irritated, feeling blue or sad, and feeling inferior to others. Each item was rated on a 5-point Likert scale ranging from 0 (not at all) to 4 (extremely). The scores of all the items were summed up to calculate a total score. For poor mental health, this study adopted a cutoff point of $\geq$10, which was consistent with previous studies and considered indicative of moderate to severe mental disorders [26]. The Chinese-version of the CBI was used to assess workers' burnout status [19], and a cutoff point of $\geq$75 defining a high level of burnout.

2.6. Data analysis

Descriptive analyses were conducted, and Chi-square and ANOVA tests were used to explore whether there were differences in sociodemographic characteristics, work-related variables, and health status among the three types of platform workers. To assess internal consistency, Cronbach's α values were examined for labor control and work conditions scales. Furthermore, principal component analysis and principal axis factor analysis with varimax rotation were employed to evaluate the construct validity of the labor control scale and the three work condition scales.

To assess predictive validity, logistic regression analysis was conducted for high burnout and poor mental health, using the labor control score as the primary explanatory variables. Models were adjusted for gender and age, as well as work conditions. Additionally, path analyses were performed to investigate whether labor control directly influenced workers' psychological distress or whether its effects were mediated through work conditions, namely job demands, job control, and work rewards. The statistical analyses were carried out using SAS version 9.4 (SAS Institute, Cary, NC, USA).

3. Results

3.1. Descriptive analysis

Table 1 and Supplementary Table S1 summarizes the sociodemographic characteristics, work status, work motives, work characteristics, and psychological distress of all platform workers and separately for each type of platform workers. More than half of the surveyed platform workers were below 40 years old (69%) and hold a college degree or above (57%).

On average, the surveyed platform workers spent 31.34 hours per week on platform work. However, there were variations across different types – ride-hailing drivers appeared to have the highest income levels and the longest working hours. The mean scores of work-related scales indicated that delivery workers reported the highest platform control, highest job demands, lowest job control, and lowest work rewards. Furthermore, delivery workers were more likely than the other two types to perceive low respect and job insecurity.

Table 3
Exploratory factor analysis of items of the scale of labor control through platforms and work condition scales: factor loadings* from the rotated factor pattern matrix with Varimax method ($n = 456$)

Scale	Items	F1	F2	F3	F4
Labor control through platforms (5)	1. Working time	0.67			
	2. Way of work	0.83			
	3. Place of work	0.80			
	4. Abide by the rules	0.59			
	5. Work performance	0.74			
Job demands (5)	1. Fast pace				0.32
	2. Hard work				0.84
	3. Physical effort				0.76
	4. Concentration				0.57
	5. Hectic work				0.72
Job control (5)	1. Learn new things			0.67	
	2. Be creative			0.78	
	3. Variety of tasks			0.69	
	4. Opinions influential		0.49	0.57	
	5. Develop abilities		0.49	0.62	
Work rewards (4)	1. Job security		0.79		
	2. Career prospects		0.77	0.32	
	3. Help and support		0.65		
	4. Appropriate income		0.54		
Variation explained		2.93	2.66	2.54	2.38
Variance explained (%)		15.40	14.01	13.37	12.54

*Only items with a factor loading ≥0.30 are listed.

Regarding safety and health risks, 14% of the surveyed workers indicated that they had suffered injured or illnesses in the past year. Moreover, 10% of the workers reported high burnout, and 22% reported having poor mental health. The prevalence of high burnout and poor mental health was highest among delivery workers compared to chore service workers and ride-hailing drivers.

3.2. Reliability and validity of the labor control scale

Cronbach's α coefficients for the labor control scale and the three work conditions scales are presented in Table 2. All coefficients were all above .81, indicating sufficient internal consistency.

From all items for the labor control scale and other scales of work characteristics, four components were extracted through principal component analysis, following Kaiser's rule with eigenvalues greater than 1 (eigenvalues = 5.14, 3.78, 2.23, and 1.15). This four-factor solution was confirmed by principal axis factor analysis (eigenvalues = 4.71, 3.31, 1.78, and .71), closely corresponding to the hypothesized factor structure. The items in the labor control scale were categorized under Factor 1, and there was no overlap in factors with items from other work condition scales. Although the fourth eigenvalue was lower than 1, the scree plot indicated the line flattened at the fifth eigenvalue (eigenvalues = .37). After varimax rotation, the four factors explained 55% of the total variance for 19 items, and their factor loadings are shown in Table 3.

3.3. Associations between labor and mental well-being

When regressed on the subordinance score in separate models that only controlled for gender and age (as shown in Table 4), workers in the highest tertile of labor control were found to have significantly greater risks for burnout and poor mental health. However, after further controlling for job demands, job control, and work rewards, the associations between labor control and psychological distress outcomes become insignificant, suggesting that these work-related factors might have mediated the influences of labor control to digital platforms. Therefore, we decided to utilize the path analysis to dismantle different routes through which labor control exert its impacts on mental health outcomes indirectly.

Table 4
Logistic regression models for burnout and poor mental health ($N = 487$)

Variable	Burnout (≧75)						Poor mental health (BSRS-5 ≧ 10)					
	Model 1			Model 2			Model 1			Model 2		
	OR	(95% CI)	p	OR	(95% CI)	p	OR	(95% CI)	p	OR	(95% CI)	p
Labor control through platforms												
Low	1			1			1			1		
Median	1.77	(0.73, 4.29)	n.s.	1.94	(0.74, 5.05)	n.s.	1.07	(0.62, 1.84)	n.s.	1.01	(0.56, 1.79)	n.s.
High	4.53	(1.80, 11.39)	**	2.32	(0.84, 6.36)	n.s.	2.47	(1.35, 4.50)	**	1.41	(0.73, 2.74)	n.s.
Job demands												
Low	—	—	—	1			—	—	—	1		
Median	—	—	—	1.69	(0.45, 6.34)	n.s.	—	—	—	1.48	(0.76, 2.86)	n.s.
High	—	—	—	6.90	(1.92, 24.79)	**	—	—	—	3.63	(1.84, 7.17)	***
Job control												
High	—	—	—	1			—	—	—	1		
Median	—	—	—	0.56	(0.24, 1.32)	n.s.	—	—	—	0.72	(0.40, 1.29)	n.s.
Low	—	—	—	0.25	(0.09, 0.69)	**	—	—	—	1.38	(0.69, 2.76)	n.s.
Work rewards												
High	—	—	—	1			—	—	—	1		
Median	—	—	—	2.41	(0.79, 7.31)	n.s.	—	—	—	2.11	(1.13, 3.95)	*
Low	—	—	—	6.93	(2.30, 20.82)	***	—	—	—	1.49	(0.75, 2.99)	n.s.

*$p < 0.05$; **$p < 0.01$; ***$p < 0.001$; BSRS, Brief symptom rating scale; n.s., not significant.
†Model 1 is adjusted for gender and age, and model 2 further adjusted for all work conditions.

Fig. 1 showed the results of path analyses for burnout and poor mental health, respectively. The findings indicated that there were significant indirect effects of labor control on workers' burnout and poor mental health, with the effects primarily mediated through higher job demands.

4. Discussion

This study evaluated the psychometric properties of a scale designed to measure the level of labor control exerted through digital platforms and explored its associations with work characteristics and workers' mental well-being. The results indicated that this scale exhibited good psychometric properties, making it suitable for distinguishing features of different types of platform work. Furthermore, higher labor control was found to be associated with greater mental health risks among platform workers, and these effect were mediated through job demands.

In terms of demographic composition, this study indicated that platform workers in Taiwan were predominantly young and well-educated individuals residing in urban areas. Our findings are consistent with previous studies from other countries [6,7,12,13,27–31]. Compared to studies from Republic of Korea and China [32,33], Taiwanese platform workers were younger and more highly educated. This observation is not surprising as platform work often demands a certain level of knowledge and digital competence. The work tenure of platform workers was predominantly less than three years. This observation may be indicative of employment instability and the relatively short history of the platform economy in Taiwan. It could also be attributed to changes in the job market after COVID-19, which might have prompted more individuals to explore work opportunities on online platforms, remote jobs, and gig economy positions. As a result, there might have been an increase in people engaging in platform work, with a significant portion having a work tenure of less than 3 years during the survey.

The impact of work conditions on psychological distress among platform workers was found to be similar to that observed in general workers, suggesting that well-established models such as the job control-demand model and effort-reward imbalance model are applicable to the platform worker context as well. Importantly, the study identified labor control through digital platforms as a significant factor influencing mental health, and this effect can be partially attributed to the increased job demands and decreased job control. When platform control is high, workers may face limited autonomy in managing their tasks, leading to decreased job control. Digital platforms may also increase job demands by exerting influence over working time and work performance evaluations. Consequently, efforts to modify labor policies and empower platform workers with greater control over their tasks and working conditions could potentially mitigate the detrimental impact of platform control on their mental health.

The findings that delivery workers experienced lower respect from others and higher risks of occupational injuries and illnesses than platform workers of the other two types deserve attention. Our findings echo previous studies, indicating that platform work characterized by higher platform subordinance may entail greater occupational safety and health risks [3,6,7].

This study represents one of the limited investigations that assess the demographic and work characteristics of the growing workforce known as platform workers. This new form of employment introduces novel challenges to work safety and health. In this context, the labor control exerted by platforms has not been thoroughly evaluated, making the scale developed in this study a valuable tool. It is worth noting that the labor control dynamics differ significantly between digital platform workers and regular employees in a corporate or industrial setting. Regular employees typically have their work schedules and patterns determined by their employers or managers. In contrast, platform workers, as independent contractors, have the freedom and flexibility to choose when and where they work. However, as shown in this study, algorithms behind platform work have replaced low- and middle-level managers, subjecting platform workers to algorithmic management [34,35]. Digital platforms use data and algorithms to continuously monitor workers' performance, assign tasks, enforce rules, and rate workers. Although these automated decisions are impersonal, the rules within the algorithms can be rigid, biased and unfair, and they often lack accountability and transparency [36,37].

However, there are certain limitations to this research that should be acknowledged.Firstly, all measurements relied on self-reported data, which may be susceptible to common method bias. Nevertheless, obtaining comprehensive register data for platform workers is challenging due to their underpresentation in formal labor registries. The use of a web-based questionnaire maximized sample recruitment, as it allowed us to employ a familiar tool for platform workers. Secondly, the convenient sampling procedure may have limited the representativeness of the study sample. Individuals who experience higher levels of social isolation and less frequent use of social media might be underrepresented in the study. As social isolation is a characteristic of many platform workers, our findings on platform control could be underestimated by omitting those with fewer social interactions. Thirdly, the cross-sectional nature of the study restricts our ability to draw causal inferences between platform subordinance, work conditions, and psychological distress. It is possible that workers experiencing higher psychological distress may be more likely to report worse work conditions. To establish stronger causality links, longitudinal studies investigating the health effects of platform work are required to replicate and validate our results.

In conclusion, while flexibility and autonomy attract workers to platform work, certain types of labor platforms exert labor control over workers' work practices and may entail significant mental health risks. Considering platform work as one form of precarious employment,it has brought challenges to existing regulatory frameworks concerning labor and social protection [38]. The role of governments in safeguarding decent work and reducing social inequalities in work quality and health should be strengthened in response to the new economic development.

Ethical considerations

The study proposal was approved by the Research Ethics Committee of National Taiwan University (202006HS0099).

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Data availability

The data that support the findings of this study are openly available in figshare at http://doi.org/10.6084/m9.figshare.24431488.

CRediT authorship contribution statement

Yawen Cheng: Writing – original draft, Data curation, Conceptualization. **Wan-Ju Cheng:** Writing – review & editing, Conceptualization. **Ro-Ting Lin:** Writing – review & editing. **Yi-Ting Wang:** Formal analysis, Data curation. **Jyh-Jer Roger Ko:** Writing – review & editing.

Conflicts of interest

All authors declare no conflicts of interest related to this study.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.shaw.2024.08.003.

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